Programming Project Phase #1

Lexical Analyzer for TINY Language

Course: CSC340 - Programming Languages and Compilation

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1.

Test Case 1: Successful Execution

1.1. Input Program

```
-- Requires 5 integer arguments, all of which are assumed to be < 100

program mytest;

n : integer;
i, j, k : integer;
x, y, z : integer;
t : array (100) of integer;

beginprogram  -- main program
  read n, i, j, x, y;
  if i < j and j <= n and i >= 0 then
     t(i) := i;
     t(j) := j;
     k := t(i);
     t(i) := t(j);
     t(j) := t(i);
  else
     while x > y or false loop
        k := (1 + y - x) * n / 3 + x;
        x := x - 10;
        write k;
     endloop;
  endif;
```

```
  write i, j, x, y, t(i), t(j);
endprogram
```

1.2. Output Tokens

```
PROGRAM
IDENT mytest
SEMICOLON
IDENT n
COLON
INTEGER
SEMICOLON
IDENT i
COMMA
IDENT j
COMMA
IDENT k
COLON
INTEGER
SEMICOLON
IDENT x
COMMA
IDENT y
COMMA
IDENT z
COLON
INTEGER
SEMICOLON
IDENT t
COLON
ARRAY
L_PAREN
NUMBER 100
R_PAREN
OF
INTEGER
SEMICOLON
```

```
BEGIN_PROGRAM
READ
IDENT n
COMMA
IDENT i
COMMA
IDENT j
COMMA
IDENT x
COMMA
IDENT y
SEMICOLON
IF
IDENT i
LT
IDENT j
AND
IDENT j
LTE
IDENT n
AND
IDENT i
GTE
NUMBER 0
THEN
IDENT t
L_PAREN
IDENT i
R_PAREN
ASSIGN
IDENT i
SEMICOLON
IDENT t
L_PAREN
IDENT j
R_PAREN
ASSIGN
```

```
IDENT j
SEMICOLON
IDENT k
ASSIGN
IDENT t
L_PAREN
IDENT i
R_PAREN
SEMICOLON
IDENT t
L_PAREN
IDENT i
R_PAREN
ASSIGN
IDENT t
L_PAREN
IDENT j
R_PAREN
SEMICOLON
IDENT t
L_PAREN
IDENT j
R_PAREN
ASSIGN
IDENT t
L_PAREN
IDENT i
R_PAREN
SEMICOLON
ELSE
WHILE
IDENT x
GT
IDENT y
OR
FALSE
LOOP
```

```
IDENT k
ASSIGN
L_PAREN
NUMBER 1
ADD
IDENT y
SUB
IDENT x
R_PAREN
MULT
IDENT n
DIV
NUMBER 3
ADD
IDENT x
SEMICOLON
IDENT x
ASSIGN
IDENT x
SUB
NUMBER 10
SEMICOLON
WRITE
IDENT k
SEMICOLON
ENDLOOP
SEMICOLON
ENDIF
SEMICOLON
WRITE
IDENT i
COMMA
IDENT j
COMMA
IDENT x
COMMA
IDENT y
```

```
COMMA
IDENT t
L_PAREN
IDENT i
R_PAREN
COMMA
IDENT t
L_PAREN
IDENT j
R_PAREN
SEMICOLON
END_PROGRAM
```

2.

Error Example 1: Unrecognized Symbol

2.1. Input Program

```
program test;
n : integer;
beginprogram
  read n;
  n := n + 1?;
  write n;
endprogram
```

2.2. Output (Screenshot Placeholder)

3.

Error Example 2: Invalid Identifier - Starts with Digit

3.1. Input Program

```
program test;
2n : integer;
beginprogram
endprogram
```

3.2. Output (Screenshot Placeholder)

4.

Error Example 3: Invalid Identifier - Ends with Underscore

4.1. Input Program

```
program test;
n_ : integer;
beginprogram
Endprogram
```

4.2. Output (Screenshot Placeholder)